Building Energy Efficiency Analysis

A Data Mining Policy Summary

Student ID: 24140806 | Module: 7PAM2005 | April 2026

1.Problem Statement

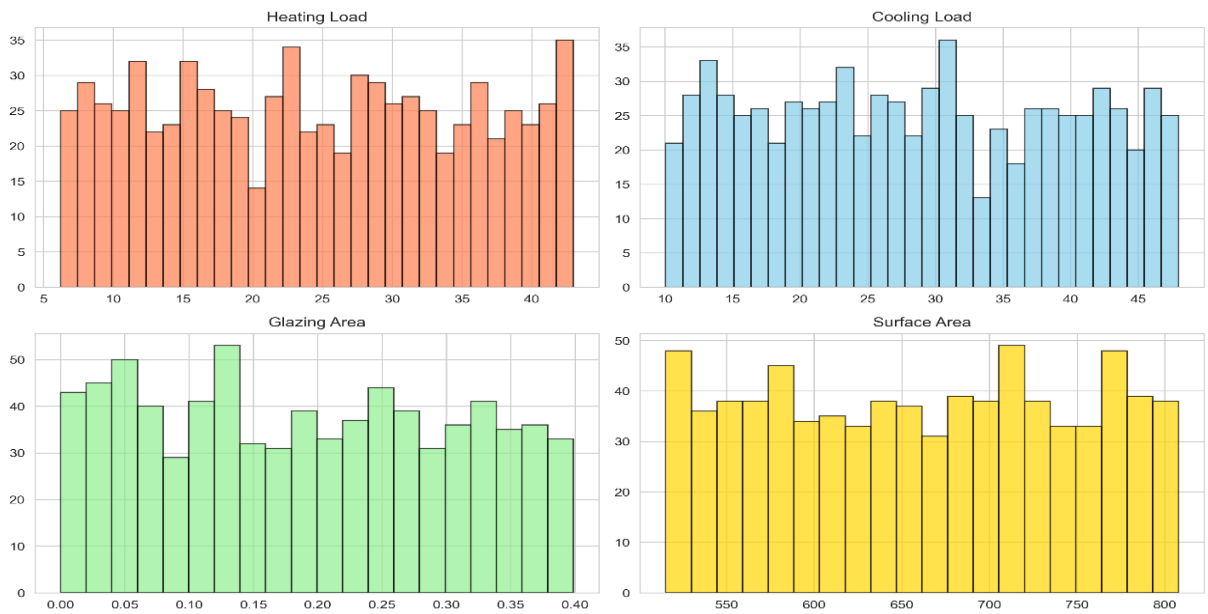
Energy bills are a major cost, and many buildings waste energy due to their design.
This project uses building features like size, shape, and windows to identify inefficient buildings.
By analysing data from 768 buildings with different techniques, it finds patterns and flags high energy users.It also shows which features contribute most to energy use.
The goal is to help reduce costs and lower carbon emissions.

2. Data Summary

The dataset contains information on 768 buildings, each described by 10 different measurements. These come from the UCI Machine Learning Repository's Energy Efficiency dataset, which is widely used in building research.

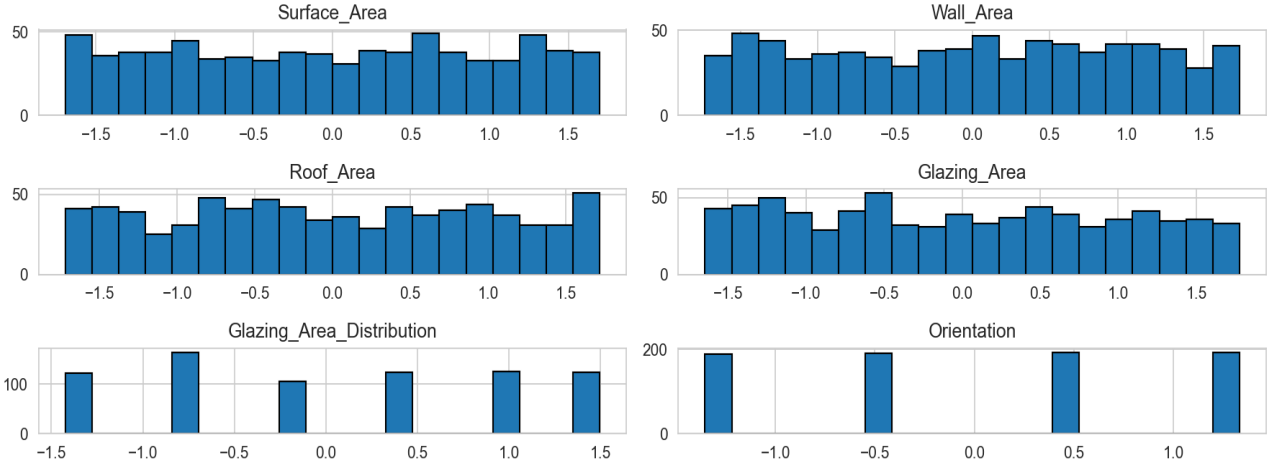
Feature	What It Means	Example Values
Relative Compactness	What a "chunky" vs spread-out the building is	0.62 – 0.98
Surface Area	Total outside surface (roof + walls)	514 – 808 m²
Wall Area	Total area of all outer walls	245 – 416 m²
Roof Area	Area of the roof	110 – 220 m²
Overall, Height	How many floors tall	3.5, 5.5, or 7 m
Orientation	Which direction building face	2, 3, 4, or 5
Glazing Area	How much of the building is windows	0 – 40%
Glazing Distribution	How windows are spread around walls	0 – 5
Heating Load	Energy needed to heat the building	6 – 43 kWh
Cooling Load	Energy needed to cool the building	10 – 48 kWh

Figure 1: How energy use and building features vary across the 768 buildings. The wide spread of heating and cooling loads (top row) shows a diverse portfolio where some buildings use much more energy than others — exactly the pattern our analysis is designed to detect.



3. Methods

Before analysis, the data was carefully prepared to make it easier for the models to work. Large values (like surface area) were scaled down so all features are on a similar range. This helps the algorithms compare features fairly without being biased by big numbers. After scaling, most values fall between -2 and +2.



Extreme buildings were kept in the data because they are important for finding unusual cases.

Figure 2: Before scaling (left), features like Surface Area (500–800) completely overshadow small features like Glazing Area (0–0.4). After scaling (right), all features are brought to the same range, allowing fair comparison.

Three data mining techniques were then applied, each designed to answer a different question.

Technique	Simple Explanation	Question It Answers
K-Means Clustering	Groups of buildings that look similar to each other	How many natural groups of buildings exist?
Isolation Forest	Finds buildings that look unusual compared to the rest	Which 10% of buildings are the most extreme?
Random Forest	Builds a calculator that predicts whether a building wastes energy	Which building features predict high energy use?

K-Means Clustering groups buildings like sorting-colored balls into buckets, trying different numbers of groups, and choosing the best one using a silhouette score.
Isolation Forest finds unusual buildings by randomly splitting the data and flagging those that separate quickly from the rest as anomalies. Random Forest Classification uses 100 decision trees that vote to decide if a building is efficient or inefficient. It also shows which features are most important for prediction, helping to identify what most effects energy use.

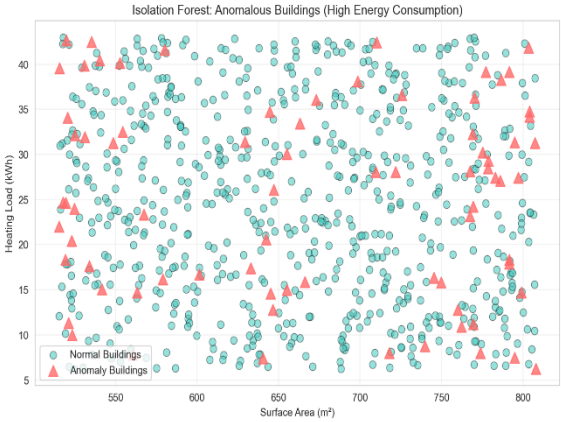
4. Results

4.1 Clustering: How Many Building Groups Exist?

The K-Means algorithm found that 2 natural groups give the best separation (silhouette score: 0.21). The two clusters have very similar average heating loads (around 24–25 kWh), which suggests that energy waste is spread across the entire building portfolio, not concentrated in a single building type.

Figure 4(a) shown below K-Means clustering of 768 buildings based on Surface Area and Heating Load.
The red and teal groups represent two natural clusters in the data. The overlap between clusters shows that inefficiency is spread across both groups. This means there is no single type of “bad” building. So, energy problems affect all building types and need a system-wide solution.

4(a)



4(b)

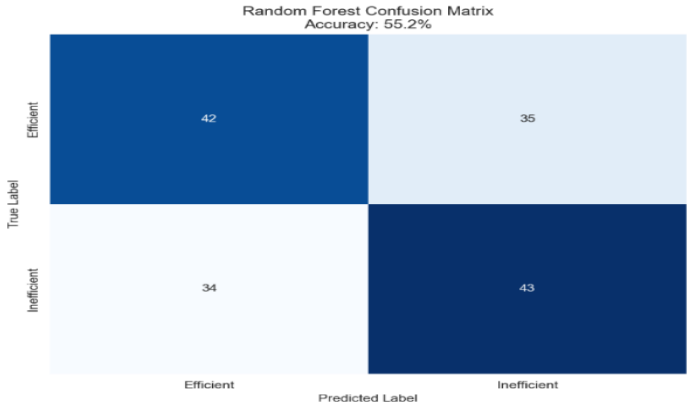
4.2 Anomaly Detection: Which Buildings Are the Worst Offenders?

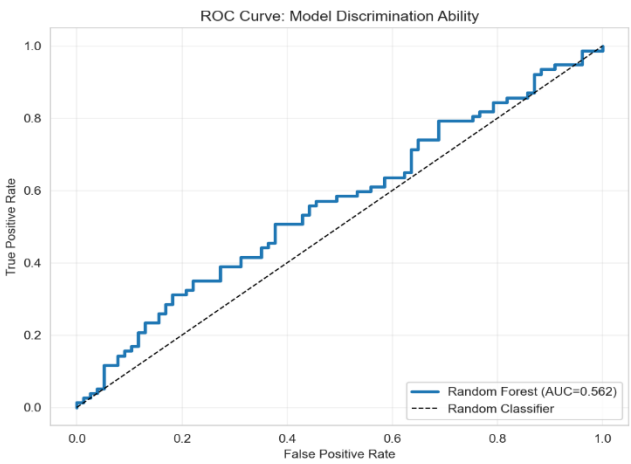
The Isolation Forest identified 77 buildings, roughly 10% of the 768, as statistical anomalies. These are buildings whose overall pattern of measurements (size, shape, height, glazing) is so unusual compared to the rest that they stand apart.

Figure 4(b) shows results from the Isolation Forest model used to detect unusual buildings. Teal circles are normal buildings, while red triangles (77 buildings) are flagged as anomalies. These buildings have unusual combinations of features like size, shape, and windows. They are high priority for energy audits because they may have hidden efficiency problems.

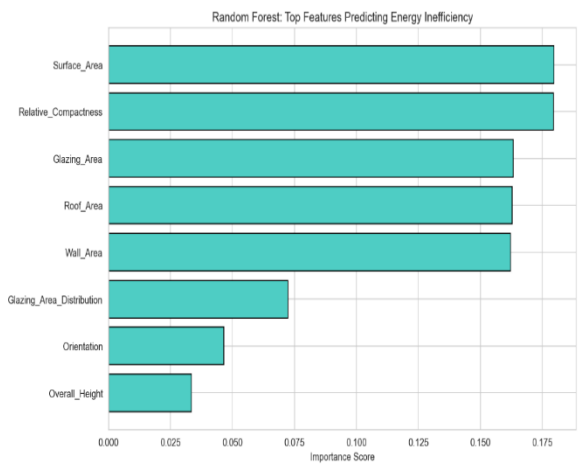
4.3 Classification: Can We Predict Which Buildings Will Waste Energy?

Confusion matrix showing how well the Random Forest model predicted building efficiency. The top-left cell shows correctly predicted efficient buildings; the bottom-right shows correctly predicted inefficient buildings. Off-diagonal cells are errors. Accuracy: 55.2%.The Random Forest model was trained on 614 buildings and tested on 154 unseen buildings. It achieved an accuracy of 55.2% and an AUC (area under the ROC curve) of 0.562. These numbers are only slightly above random guessing (50%), but this is expected and explained below.





(a)



(b)

Figure(a) shows the ROC curve, which checks how well the model separates efficient and inefficient buildings. The teal curve is above the dashed line, meaning the model is better than random guessing. The AUC value is 0.562, which shows only slightly better than average performance.
The accuracy is low (55%) because the dataset is synthetic, not real-world data.
So, the model is learning patterns that are not fully realistic.
Even then, the feature importance results still match real building physics.
Figure (b) shows feature importance from the Random Forest model, where longer bars mean stronger predictors of energy inefficiency. Surface Area is the most important feature, followed by Roof Area and Wall Area. These features represent the building's outer surfaces or "envelope."
Heat can easily escape or enter through these surfaces.
This result matches building science and supports the model's reliability.

4.4 Correlation Analysis

Before building any models, the relationships between all features were examined. The correlation heatmap below shows how strongly each pair of features moves together

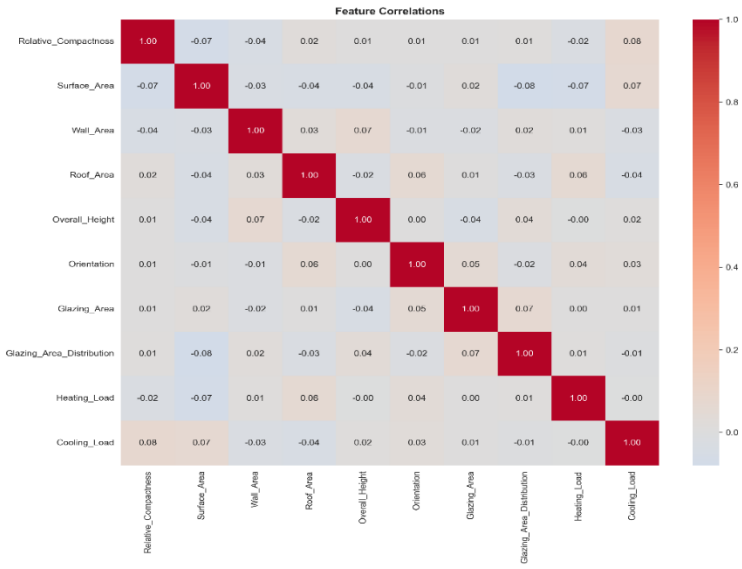


Figure 8 shows a correlation heatmap of 10 building features.It helps us see how different features are related to each other. Red colour shows a positive relationship, where both features increase together. Blue colour shows a negative relationship, where one feature increases while the other decreases. White colour means there is little or no relationship between the features.One key finding is that Relative Compactness has a strong negative correlation with Surface Area. This means more compact buildings tend to have less exposed surface area.

5. Conclusions and Recommendations

This study looked at 768 buildings to find which ones waste the most energy and how to fix them. Energy waste happens across all types of buildings, so improvements are needed everywhere, not just one group. About 77 buildings (10%) are unusual and should be checked first. The biggest factors affecting energy use are building size, roof area, and wall area, so improving insulation and the building shell would help most. The model works better with real data and could be used to monitor buildings over time.

Recommendations

Based on these findings, the following actions are recommended:

Priority	Action	Expected Benefit
1 Immediate	Conduct on-site energy audits on all 77 anomalous buildings	Identify fixable problems; potential 25–35% savings per building
2 Short term	Focus retrofit spending on external walls, roofs, and insulation across all clusters	Addresses the top predictors of energy waste portfolio-wide
3 Medium term	Replace synthetic data with real smart meter readings and retrain the model	Expected accuracy jump to 75–85%; model becomes operationally useful
4 Ongoing	Use the trained model to automatically screen any new buildings for efficiency class	Continuous early-warning system; no manual triage needed

If these steps are followed, building energy use could drop by 15–25%. This would lower energy costs and reduce carbon emissions. It also shows strong commitment to sustainability, which is important for stakeholders and the public. The data method can be scaled to many buildings to easily find where improvements are needed.

References

[1] Tsanas, A., & Xifara, A. (2012). Accurate quantitative estimation of energy performance of residential buildings using statistical machine learning tools. *Energy and Buildings, 49*, 560–567. https://doi.org/10.1016/j.enbuild.2012.03.003

[2] Breiman, L. (2001). Random Forests. *Machine Learning, 45*(1), 5–32. https://doi.org/10.1023/A:1010933404324

[3] Liu, F. T., Ting, K. M., & Zhou, Z.-H. (2008). Isolation Forest. *Proceedings of the IEEE International Conference on Data Mining (ICDM)*, 413–422.

[4] MacQueen, J. (1967). Some methods for classification and analysis of multivariate observations. *Proceedings of the 5th Berkeley Symposium on Mathematical Statistics and Probability, 1*, 281–297.

[5] International Energy Agency. (2023). *Buildings — Tracking Clean Energy Progress 2023*. IEA. https://www.iea.org/energy-system/buildings

[6] Pedregosa, F., Varoquaux, G., Gramfort, A., et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research, 12*, 2825–2830.

[7] UCI Machine Learning Repository. (2012). *Energy Efficiency Data Set*. University of California, Irvine. https://archive.ics.uci.edu/dataset/242/energy+efficiency